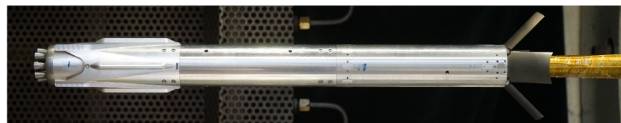


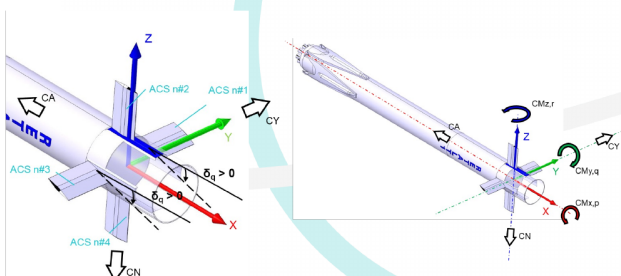
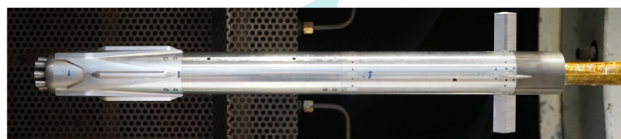
RETALT1 demo problem

Predict aerodynamic loads for a control-surface configuration the model has never seen

Petals

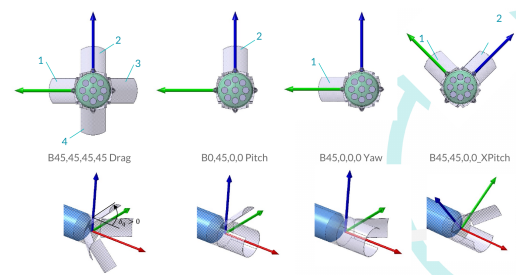
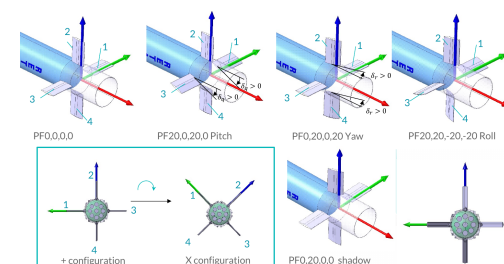


Planar Fins



CFSE - I. Vuc D. Charbonnier - August 27, 2022

RETALT



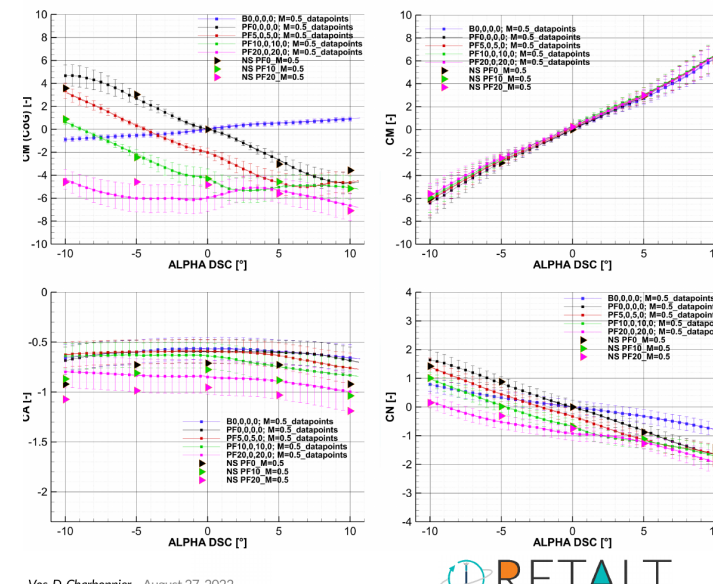
Inputs: Mach, angle of attack, surface family and layout, and four surface deflections. **Outputs:** three force coefficients and five moment coefficients.

The benchmark asks for generalization across **entire unseen control-surface configurations** - not a CFD field and not nearby-row interpolation.

The data is 170 polars - not 6,908 independent rows

Adjacent angle-of-attack samples belong to the same processed aerodynamic curve

- Data extracted at uprunning alpha-polar
- Hysteresis Data partly included
- Data extracted generally in $-10^\circ < \alpha_{psc} < 10^\circ$ in steps of 0.5°
 - Exceptions if measurement data not available or due to hysteresis
- Data was acquired at higher frequencies, resampled to 1000 Hz and filtered with low pas filter with 3Hz cutoff frequency
- Data modifications:
 - For configurations symmetric in the beta plane for CN, CM_y, CM_y(CoG) the data was shifted to be exactly 0 in $\alpha = 0^\circ$
 - For the "drag"-configurations (PF0,0,0,0) The data at negative α_{psc} was mirrored from the values at positive α (as the measured α range for the positive values was larger)
 - In CFD Data of UF1 and UF3, values at $\alpha = 0^\circ$ directly set to 0 for symmetric configurations
- In the following:
 - Lines represent filtered measurement data
 - Dots represent AEDB data points
 - Triangles represent Navier-Stokes computations
 - Squares represent Low Resolution Euler computations
 - Orange contour: X configuration



TRAIN

4,408 rows - 109 curves

VISIBLE VALIDATION

779 rows - 19 complete Mach 2.5 curves

SEALED EVALUATION

1,721 rows - 42 curves - 7 unseen configurations

Never random-split neighboring angle-of-attack rows. Hold out **whole Mach curves** and test on **whole configurations**.